Economia das Organizações e dos Incentivos — MPE Insper

Exercise List: Lecture 1: Dominance and Iterated Elimination (graduate level)

Instructions. Multiple-choice items: mark the single correct option (no penalty for errors). *True/False* items: every 2 wrong items cancel 1 correct one (score = $\max\{0, \text{correct} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step; a result the item asks you to derive may not be quoted. All questions are symbolic: state the admissible range of every parameter you use. Throughout, "dominated" means dominated by another *pure* strategy, as in Lecture 1; "IESDS" removes, at every round, every strictly dominated strategy of every player. Lecture 2 material (Nash equilibrium, mixed strategies) is neither needed nor allowed.

Block 1 — Normal form, dominance and social dilemmas *(Lecture 1 • 25 points)*

Question 1. (10 points) Objective items.

a) (3 points) *Dominance × Pareto optimality.* A medical clinic in Recife is owned by $n \geq 2$ partners. Each partner simultaneously either exerts effort (E, private cost $c > 0$) or shirks (S). If k partners exert effort, the clinic's net revenue is $Y(k) = yk + \theta\binom{k}{2}$, with $y > 0$ and $\theta \geq 0$ (every pair of active partners adds θ through cross-referrals), and revenue is split equally. The set of costs c for which shirking is strictly dominant for every partner *and* the resulting profile is Pareto dominated by some other profile is:

A. $(y/n,\ y)$

B. $((y + \theta(n-1))/n,\ y + \theta(n-1)/2)$

C. $((y + \theta(n-1))/n,\ y + \theta(n-1))$

D. $(y/n,\ y + \theta(n-1)/2)$

E. $((y + \theta)/n,\ y + \theta n/2)$

b) (3 points) *Weak dominance × IESDS.* A court-ordered sale (*leilão judicial*) of a warehouse in Campinas uses a sealed-bid second-price rule with a public reserve price $r > 0$: $n \geq 2$ bidders submit bids $b_j \in [0, \infty)$; if every bid is below r there is no sale; otherwise the highest bidder wins (ties broken uniformly at random) and pays $\max\{r, \text{highest other bid}\}$. Bidder i values the warehouse at $v_i > r$. Which statement is correct?

A. Bidding v_i strictly dominates every other bid, so IESDS selects truthful bidding for every bidder whose value exceeds r.

B. Bidding v_i weakly dominates the bids above v_i only; bids in $[r, v_i)$ are not weakly dominated, since they lower the price paid.

C. Bidding $v_i - r$ weakly dominates every other bid, because the reserve price is a cost that the winner must net out of the value.

D. Bidding v_i weakly dominates every other bid, yet no bid of any bidder is strictly dominated, so IESDS removes no strategy at all.

E. Bidding v_i weakly dominates every other bid, and the bids below r are strictly dominated by v_i, so IESDS removes them.

c) (2 points, T or F) *Invariance × dominance.* Let G be a finite game with a strict dominant-strategy equilibrium s^D. Then there is $\varepsilon > 0$ such that s^D is a strict dominant-strategy equilibrium of every game G' with the same players and strategy sets whose payoffs satisfy $|u_i'(s) - u_i(s)| < \varepsilon$ for every i and every s.

d) (2 points, T or F) *Pareto optimality × dominance.* In every finite game, with any number of players, that has a strict dominant-strategy equilibrium s^D, every profile that Pareto dominates s^D has *every* player i using a strategy different from s_i^D.

Question 2. (15 points) *Dominance × Pareto optimality and social dilemmas × solution-concept criteria.* An accounting firm in Belo Horizonte is a partnership of $n \geq 2$ partners. Simultaneously,

each partner i chooses an effort level $e_i \geq 0$. Firm revenue is $y \sum_j e_j$ $(y > 0)$; partner i bears the private cost $\frac{c}{2} e_i^2$ $(c > 0)$ and receives the share $\alpha_i \geq 0$ of revenue, with $\sum_i \alpha_i = 1$ fixed in the partnership agreement. Efforts are not observable by the other partners, so nothing else can be contracted on; each partner cares only about her own payoff. Unless stated otherwise, $\alpha_i = 1/n$.
a) (3 points) Write the game in normal form $G = (I, \{S_i\}, \{u_i\})$. Prove that every partner has a strictly dominant strategy, for any shares (α_i), and find it. Draw, in the (e_i, u_i) plane, partner i's payoff as a function of her own effort for two different values of the others' total effort, and read the dominance property off the figure.
Resolution:

b) (3 points) Take equal shares. Show that the game has a unique strict dominant-strategy equilibrium and a unique iterated-elimination equilibrium. Prove that this profile is Pareto dominated, and find the set of symmetric effort levels $\bar{e}$ such that $(\bar{e}, \ldots, \bar{e})$ Pareto dominates it. Which symmetric profile is best for every partner?
Resolution:

c) (3 points) Return to general shares. Prove that no sharing rule $(\alpha_i \geq 0, \sum_i \alpha_i = 1)$ makes the efficient profile $(y/c, \ldots, y/c)$ a dominant-strategy equilibrium. Characterize the sharing rule that maximizes total surplus $W \equiv \sum_i u_i$ at the dominant-strategy equilibrium, and the rule that minimizes it.
Resolution:

d) (3 points) With equal shares, quantify the social dilemma: total surplus at the efficient profile, W^*, and at the dominant-strategy equilibrium, W^D; the relative loss L/W^*, where $L \equiv W^* - W^D$;

the payoff gap per partner; and the effort ratio e^D/e^*.
Resolution:

e) (3 points) Parameter analysis. Prove the sign of the effect of y, c and n (treated as continuous) on e^D, e^*, the per-partner payoff u^D, total surplus W^D and the relative loss L/W^*. Summarize in a table, study $n \to 1$, $n \to \infty$ and $c \to 0$, and give the economic reading.
Resolution:

Block 2 — Iterated elimination of strictly dominated strategies *(Lecture 1 • 25 points)*

Question 3. (10 points) Objective items.
a) (3 points) *IESDS × common knowledge of rationality.* A distributor (player 1, rows) and a retailer (player 2, columns) in the Zona Franca de Manaus play the game below, in which $x \in \mathbb{R}$ is the distributor's payoff at (U, C):

	L	C	R
U	$2, 3$	$x, 2$	$5, 0$
M	$5, 3$	$4, 4$	$2, 2$
D	$1, 4$	$1, 5$	$4, 6$

IESDS selects a unique profile if and only if, and the number of rounds it takes is:

A. $x > 1$; four rounds

B. $x > 1$; three rounds

C. $1 < x < 4$; three rounds

D. $x < 4$; two rounds

E. $1 < x < 4$; four rounds

b) (3 points) *Weak dominance × order of elimination.* In the game below, consider iterated elimination of *weakly* dominated strategies, removing one strategy at a time in any order until no

strategy is weakly dominated:

	L	C	R
U	$1,2$	$1,0$	$0,1$
M	$1,2$	$2,2$	$2,1$
D	$0,0$	$2,1$	$1,2$

Which statement is correct?

A. Some orders end at the single profile (M, L) and others at the single profile (M, C), so player 1's final payoff (1 or 2) depends on the order.

B. Every order ends at (M, C), because M weakly dominates both U and D, after which player 2 prefers C.

C. IESDS (strict) already selects (M, C), and every order of weak elimination agrees with it.

D. Removing all weakly dominated strategies at once in every round ends at the single profile (M, C).

E. Some orders end at (M, L) and others at (U, L), so player 1's final payoff is 1 in every order.

c) (2 points, T or F) *IESDS* × *existence.* In every normal-form game, IESDS leaves every player at least one strategy: $S_i^\infty \neq \varnothing$ for every i.

d) (2 points, T or F) *IESDS* × *order of elimination.* In a finite game, removing at each step *some* (not necessarily all) of the strictly dominated strategies, in any order, until none is left always ends at the same sets S_i^∞ as the lecture's procedure.

Question 4. (15 points) *IESDS* × *common knowledge of rationality* × *solution-concept criteria (uniqueness, invariance).* Two cement producers supply a regional market in the North of Brazil and choose quantities simultaneously. Inverse demand is $P(Q) = \max\{a - bQ, 0\}$ with $Q = q_1 + q_2$; firm i has constant marginal cost $c_i \in (0, a)$ and no fixed cost; $S_i = [0, \infty)$ and $u_i = \pi_i(q_i, q_j) = q_i(P(q_i + q_j) - c_i)$.

a) (3 points) Show that, for each $q_j \geq 0$, $\pi_i(\cdot, q_j)$ is strictly increasing up to a unique profit-maximizing quantity $\mathrm{BR}_i(q_j)$ and strictly decreasing beyond it, and find BR_i. Determine S_i^1 exactly, and say where the assumption $c_i > 0$ is used.

Resolution:

b) (3 points) Show that every S_i^k is a closed interval $[L_i^k, U_i^k]$ and derive the recursion linking its end points to those of S_j^{k-1}. Interpret round k in terms of knowledge of rationality.

Resolution:

c) (3 points) Characterize S^∞: prove that it is a single profile for every admissible (a, b, c_1, c_2), find it in closed form (including the case in which a firm is driven out), and show that, in the interior case, no finite round leaves a single profile. Draw, in the (q_1, q_2) plane, the two functions BR_1, BR_2 and the sets S^1, S^2, S^3, for $c_1 < c_2$.
Resolution:

d) (3 points) Now $n \geq 3$ identical firms, each with marginal cost $c \in (0, a)$. Characterize S^∞ and quantify the range of industry output and of the market price compatible with IESDS; compare with $n = 2$.
Resolution:

e) (3 points) Parameter analysis for the interior duopoly. Prove the sign of the effect of a, b, c_1, c_2 on q_1^{ES}, industry output Q^{ES}, the price P^{ES} and firm 1's profit π_1^{ES}. Tabulate, study $c_1 \to c_2$, $c_1 \uparrow (a + c_2)/2$ and the move from $n = 2$ to $n = 3$, and read the results against the uniqueness and invariance criteria.
Resolution:

Block 3 — Integrative: spatial competition *(Lecture 1 • 25 points)*

Common setting for Questions 5 and 6. Along a stretch of highway in the interior of São Paulo state, $n \geq 2$ towns are equally spaced (distance 1 between neighbours) and numbered $1, \ldots, n$; unless stated otherwise each town has one unit of households. Two fuel-retail chains, A and B, simultaneously choose one town each in which to open a single station (both may choose the same

town). Prices and products are identical and fixed. Every household refuels at the nearest station, measured along the highway, and splits evenly if both are equally distant. Each chain maximizes the number of households it serves. A household's travel cost is its distance to the station it uses; total travel distance is the sum over households.

Question 5. (10 points) Objective items.
a) (3 points) *Normal form × IESDS × welfare.* Suppose instead that the $n \geq 3$ towns are equally spaced around a ring road (like the Rodoanel around São Paulo), with distance measured along the ring. Which statement is correct?

A. A chain serves more households the farther it is from its rival, so IESDS selects antipodal pairs, which also minimize total travel distance.

B. As on a straight highway, IESDS makes the chains cluster in a single town, a profile that maximizes total travel distance.

C. Every profile gives each chain $n/2$ households, IESDS eliminates nothing, and an antipodal pair of stations minimizes total travel distance.

D. Every profile gives each chain $n/2$ households, IESDS eliminates nothing, and total travel distance is the same in every profile.

E. Every profile gives each chain $n/2$ households, so every town is a strictly dominant strategy and every profile is a strict dominant-strategy equilibrium.

b) (3 points) *IESDS × non-uniform population.* Back on the straight highway, let n be odd and let town 1, a regional capital, have $1 + 2h$ households, with $h \in \{1, \ldots, (n-1)/2\}$; every other town keeps one. IESDS selects:

A. the median town $(n+1)/2$ for both chains, since IESDS depends only on the number of towns

B. town $(n+1)/2 - h$ for both chains

C. town $(n+1)/2 - 2h$ for both chains

D. towns $(n+1)/2 - h$ and $(n+1)/2 - h + 1$ for each chain

E. town 1, the capital, for both chains, for every $h \geq 1$

c) (2 points, T or F) *Dominance × IESDS.* With uniform population and any odd $n \geq 3$, the median town is a strictly dominant strategy for each chain, which is why IESDS selects it.

d) (2 points, T or F) *Dominance × policy.* Let $n \geq 4$ be even. A municipal licensing rule allows chain A only in towns $1, \ldots, n/2$ and chain B only in towns $n/2 + 1, \ldots, n$. Then each chain has a strictly dominant strategy, and the two stations end up in adjacent towns at the centre of the highway.

Question 6. (15 points) *Integrative: normal form × dominance and IESDS × Pareto optimality × solution-concept criteria (existence, uniqueness, invariance).* Use the common setting above, with uniform population.
a) (3 points) Write the game in normal form for a general n, with each chain's payoff in closed form, and prove that it is constant-sum. Deduce which location profiles are Pareto optimal for the two chains. Who, then, can be harmed by the chains' choices?
Resolution:

b) (3 points) Characterize S^∞ for every $n \geq 2$: which towns are eliminated in each round, how many rounds there are, and which profiles survive, separately for odd and even n. Prove your characterization. What order of knowledge of rationality does the prediction require?
Resolution:

c) (3 points) Replace the towns by a continuum: households are uniformly distributed on $[0, 1]$ (unit mass) and each chain locates anywhere in $[0, 1]$. Characterize S^∞. Compare with b) as the spacing between towns shrinks, and evaluate IESDS against the existence, uniqueness and invariance criteria.
Resolution:

d) (3 points) Welfare. Find the minimum total travel distance $D^{\min}(n)$ over all location pairs and the total travel distance $D^{ES}(n)$ at the IESDS survivors. Prove that $D^{ES}/D^{\min} = 2$ for every odd $n \geq 3$, and characterize the ratio for even n. For the continuum version, draw each household's travel distance as a function of its location when (i) both stations are at $\frac{1}{2}$ and (ii) the stations are at $\frac{1}{4}$ and $\frac{3}{4}$, and read the totals off the figure.
Resolution:

e) (3 points) Parameter analysis. Let towns be $d > 0$ apart and hold $N > 0$ households each. Prove how n, d and N affect the number of IESDS rounds K, the IESDS and minimum travel distance per household, and their ratio. Tabulate, study $n = 2$ and $n \to \infty$ (in units of the stretch length $d(n-1)$), and give the economic and policy reading.
Resolution: